Chapter 4 Mobility Management

Lecture Mobile Communication

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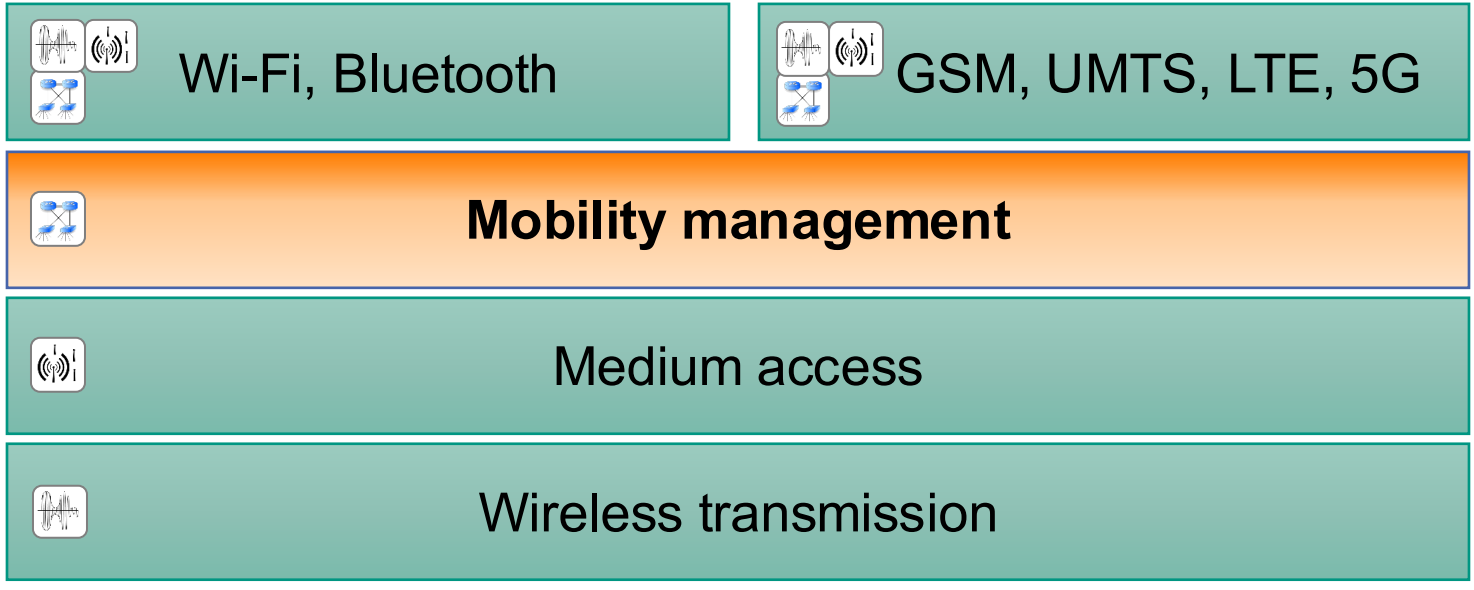
Structure


Wi-Fi, Bluetooth
GSM, UMTS, LTE, 5G
Mobility management
Medium access
Wireless transmission

General Thoughts on Mobility Management

Motivation

- Basic problem: We want to establish and maintain the connection to a mobile station

- Necessary prerequisites
  - Position management: Location (network access point) must be known
    - Base station in cellular network, access point in WLAN
  - Handover: When leaving the coverage area of an access point, a connection to a new access point must be established
  - (Re-)routing: Connections / data packets must be routed to new access point

Position Management

Position Management

- Goal: Establish a connection to a mobile device at an unknown location

- Basic Operations
  - Update: Triggered by the device to communicate its current (approximate) position to the communication system
  - Search: Triggered by the communication system to determine the (unknown) position of a device; commonly denoted as paging

- Observation: Updates increase search efficiency
  - Without position information, you have to search for a device in the entire system
  - Paging can be very time-consuming! No effort for updating, but a lot of effort for searching

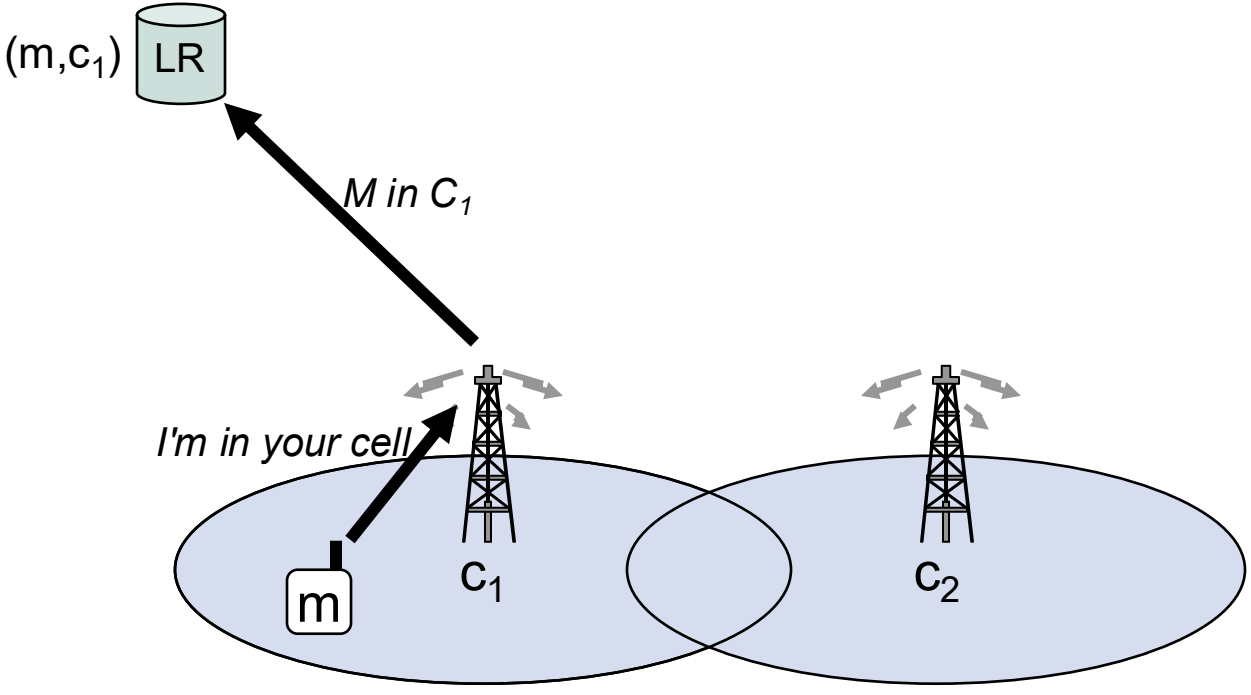
Position Management: Basic Approach

- Assumptions
  - Position granularity: Single Cell operated by a base station in a cellular network
  - Central Database: Location Register, (LR)
  - For each mobile device, LR stores the most recent known cell
    - I.e. mapping (device, cell identifier)

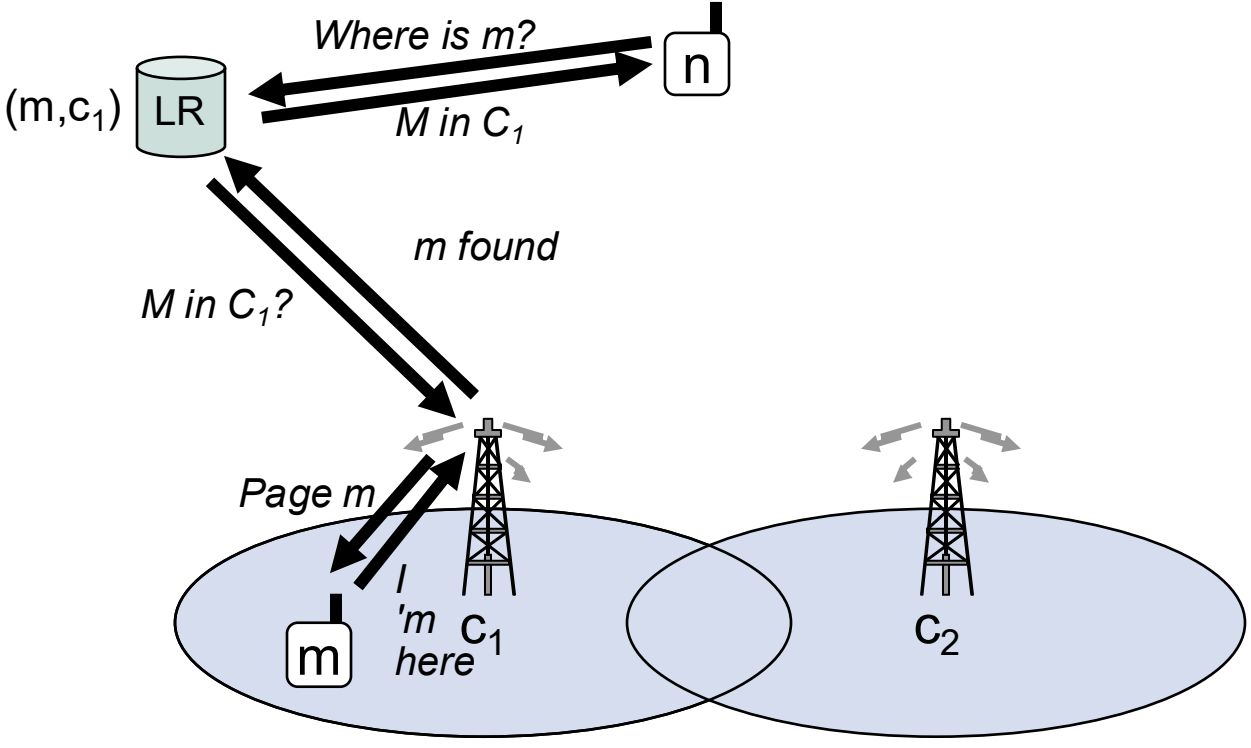
- Protocol
  - Mobile device notifies LR of current cell when it is turned on
  - With each cell change, device informs the LR (update)
  - LR is used to page the device (search)
    - LR contacts access point in current cell
    - Access point sends broadcast
    - Mobile device responds (when turned on)

- Outdated LR entries can be handled by soft state approach
  - Each entry has a limited lifetime and must be renewed before it expires
  - Can dramatically increase refresh rate

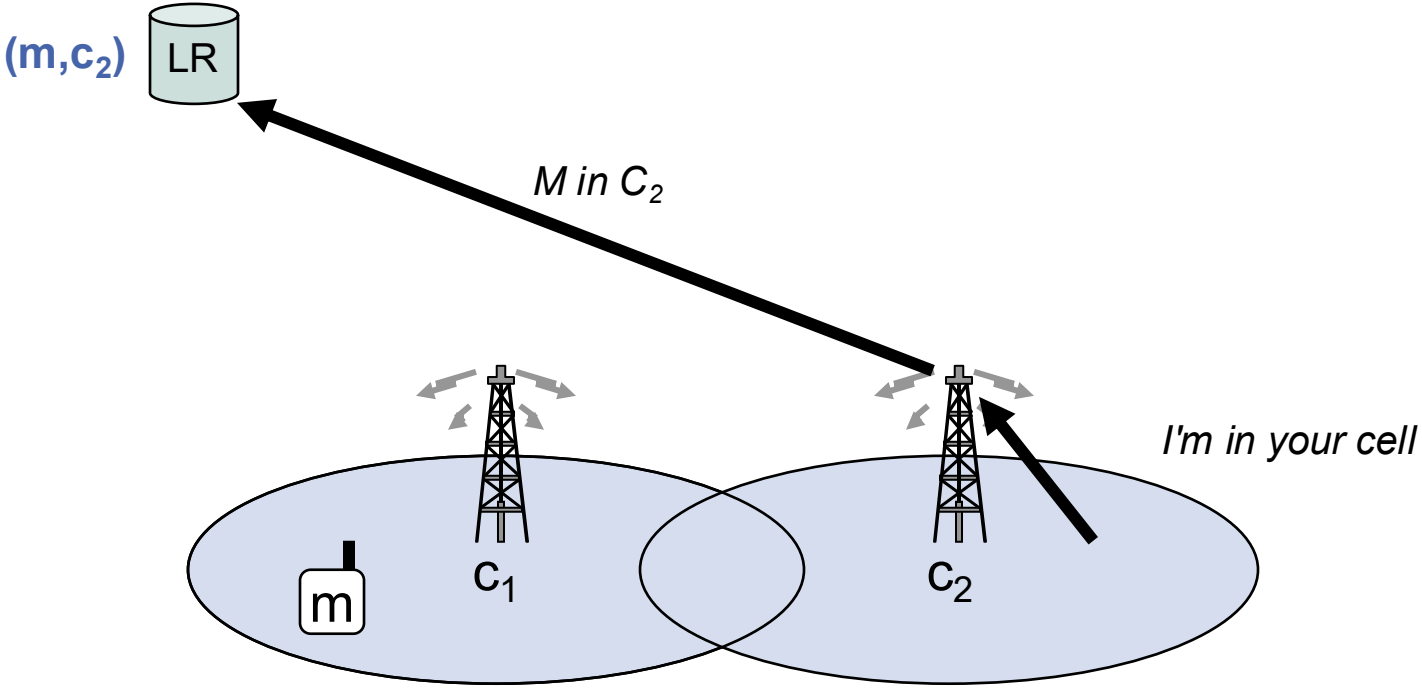
Basic Approach: Initial Update



Basic Approach: Paging



Basic Approach: Updating



Updates vs. Paging (1)

- Instead of constantly updating positions, why not just page in all cells of the network?

- What's bad about updates?
  - Sending updates drains mobile device batteries
  - Waste of bandwidth in the uplink
  - Additional load on the infrastructure (network and databases)

- What's bad about paging?
  - Waste of bandwidth in the downlink for requests
  - Waste of energy and bandwidth in the uplink for replies
  - Connection establishment / data transfer delayed
  - Additional load on the infrastructure (network)

Updates vs. Paging (2)

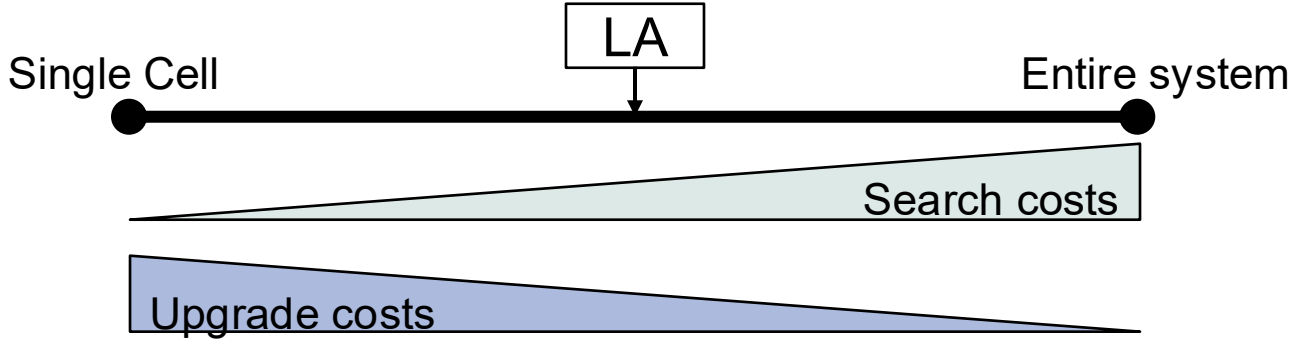
- Paging only within a single cell
  - Potentially large number of updates (for each change of a device to another cell)
  - In cellular networks, typical users change cells every 30 – 40 seconds (i.e. 1,000 cell changes per 8h day)

- Optimization: Location Area (LA) consisting of several cells
  - Lower update rates: Only when device moves to new positioning area
  - Paging required within all cells in a location area

- Location area size has a major impact on system performance

Using Location Areas

- Principle
  - The coverage area of the network is divided into a set of location areas (LAs)
  - Each location area consists of a set of spatially connected cells
  - LR manages mapping (Device, LA)
  - Update: When changing LA instead of changing cells
    - Reduces refresh rate
  - Search: Paging in all cells within LA
    - Increases effort for paging



Basic Approach: Replication/Splitting

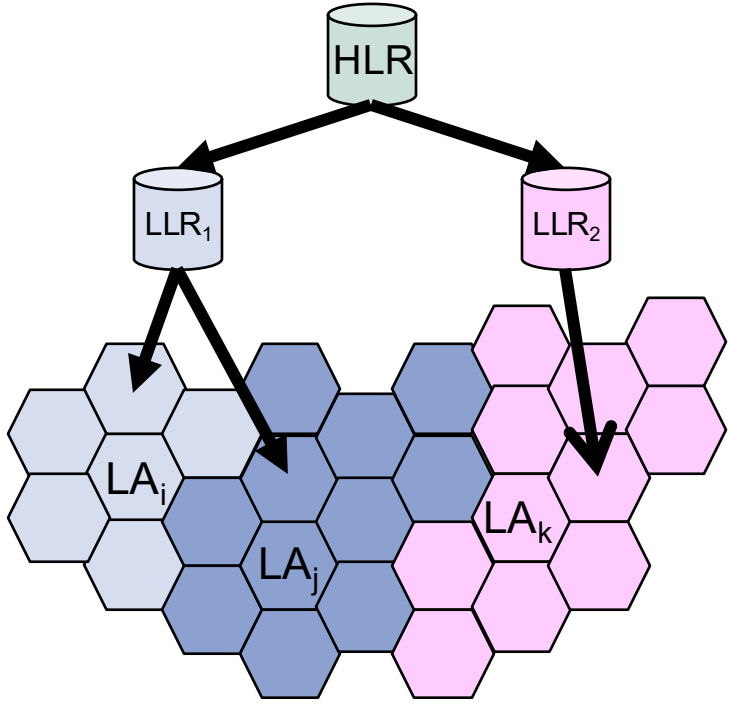
- Obvious problems and solutions
  - LR is a single point of failure
    - Possible solution: Replication of LR functionality
  - LR is bottleneck in terms of system performance
    - Possible solution: Distribution of responsibilities among different LR entities

- Design issues for distribution of responsibilities
  - How are responsibilities distributed?
  - How are the position registers organized (e.g. flat, hierarchical)?
  - Where in the system should position registers be placed?

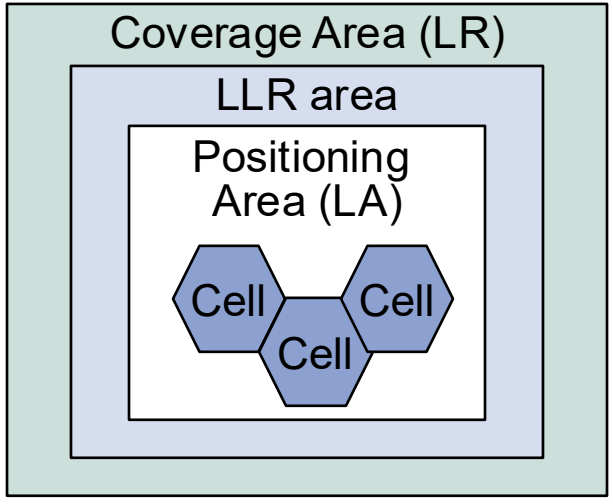
Location Register Hierachy

- Problem: LR typically far away, making updating time-consuming

- Approach
  - Introduction of a level of local location registers (LLR)
  - LLR responsible for the number of spatially connected LAs



Register Hierarchy



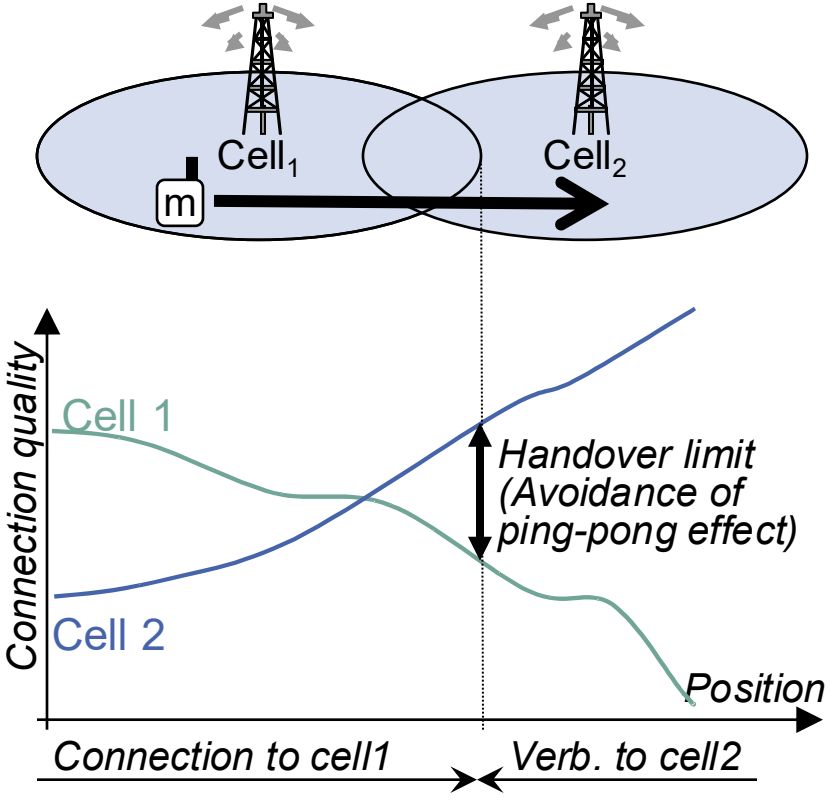
- LLR manages mapping (device, LA) for each device in its LLR scope
- LR manages mapping (device, LLR) for the LLR responsible for a device

Handover

Handover (1)

- Problem: Cell change during ongoing communication

- Reasons for connection handover
  - Wireless connection quality deteriorates
  - Communication via other cells requires fewer radio resources
  - Supported range exceeded
  - Current cell overloaded
  - Maintenance



Handover (2)

- Goal: Establish a wireless connection between the device and the new access point

- Basic approaches
  - Mobile-initiated handover
    - E.G. WLAN
  - Network-initiated handover
    - E.g. GSM (in this case mobile-assisted, i.e. mobile station supports network through measured values)

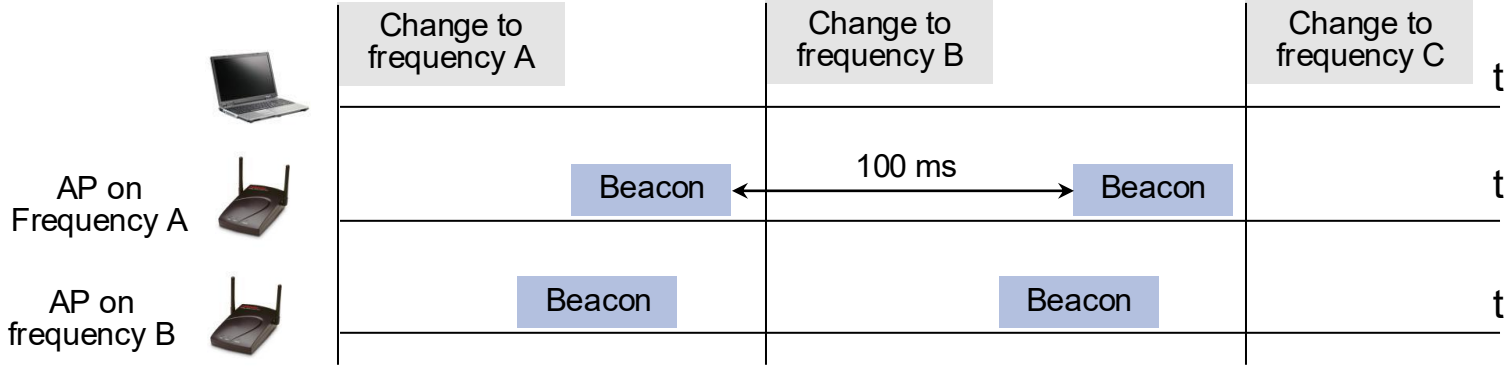
- Example: Handover between WLAN access points (AP)
  - The handover will take place in four phases
    - Determination of the loss of radio connection to the old AP
    - Search for new AP
    - *(Re-)Authentication with new AP*
    - *(Re-)Association with new AP*

Detection of Connection Loss

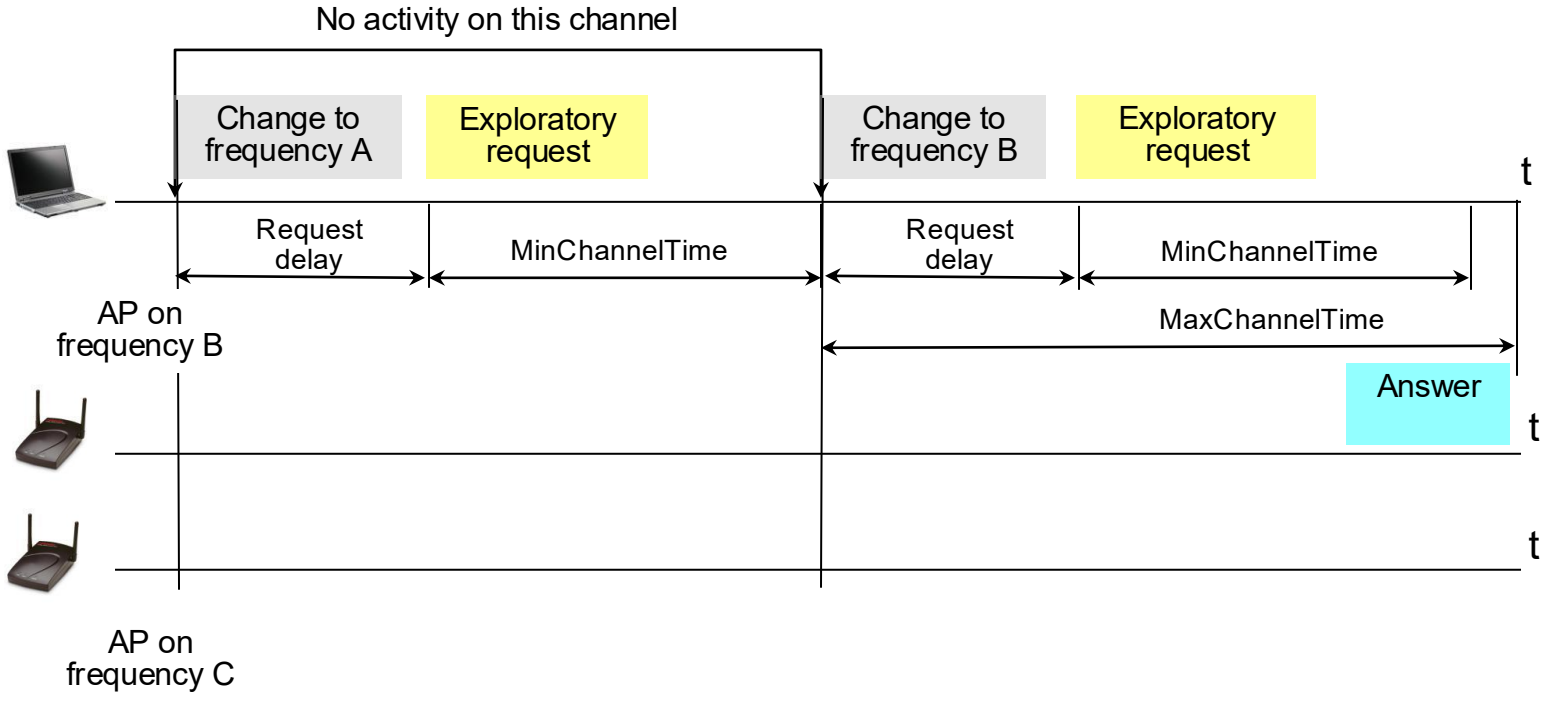
- Low signal strength
  - Most common approach
  - Handover when signal strength below threshold

- Incorrect transmission of radio frames
  - E.g., checksum failure
  - Slower than signal strength measurement
  - May be caused by other reasons
    - Collision
    - Fast fading

Discovery of new AP : Passive Scanning



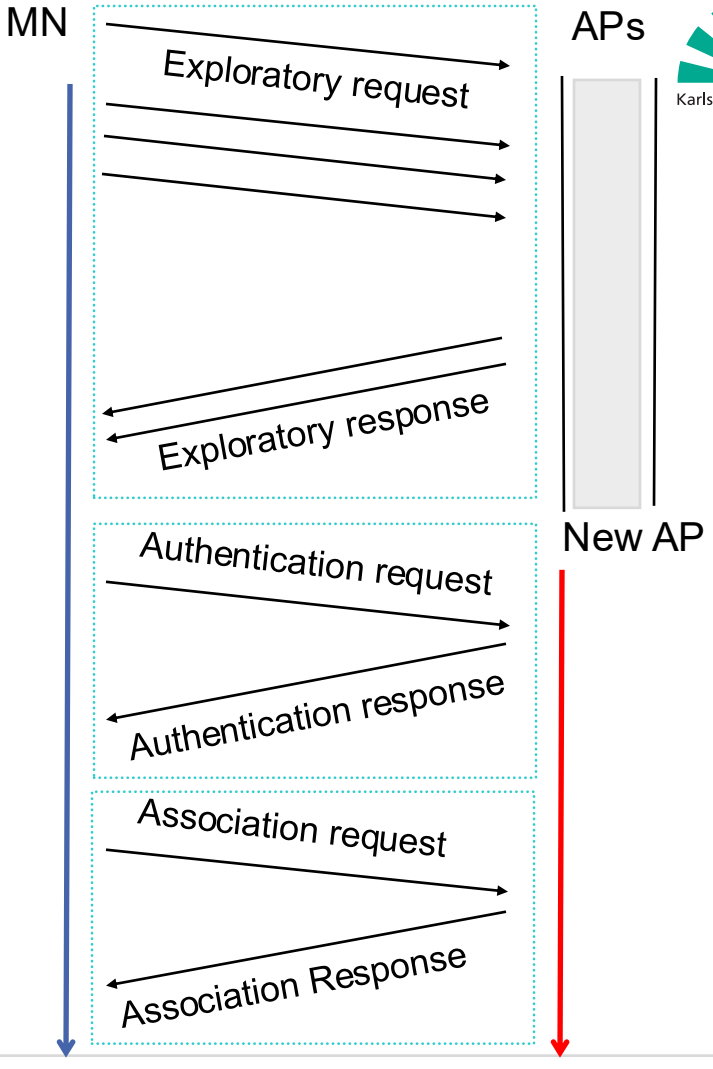
Discovery of new AP: Active Scanning



Handover Execution

- Large variation with the same hardware and configuration due to stochastic behavior of the radio interface

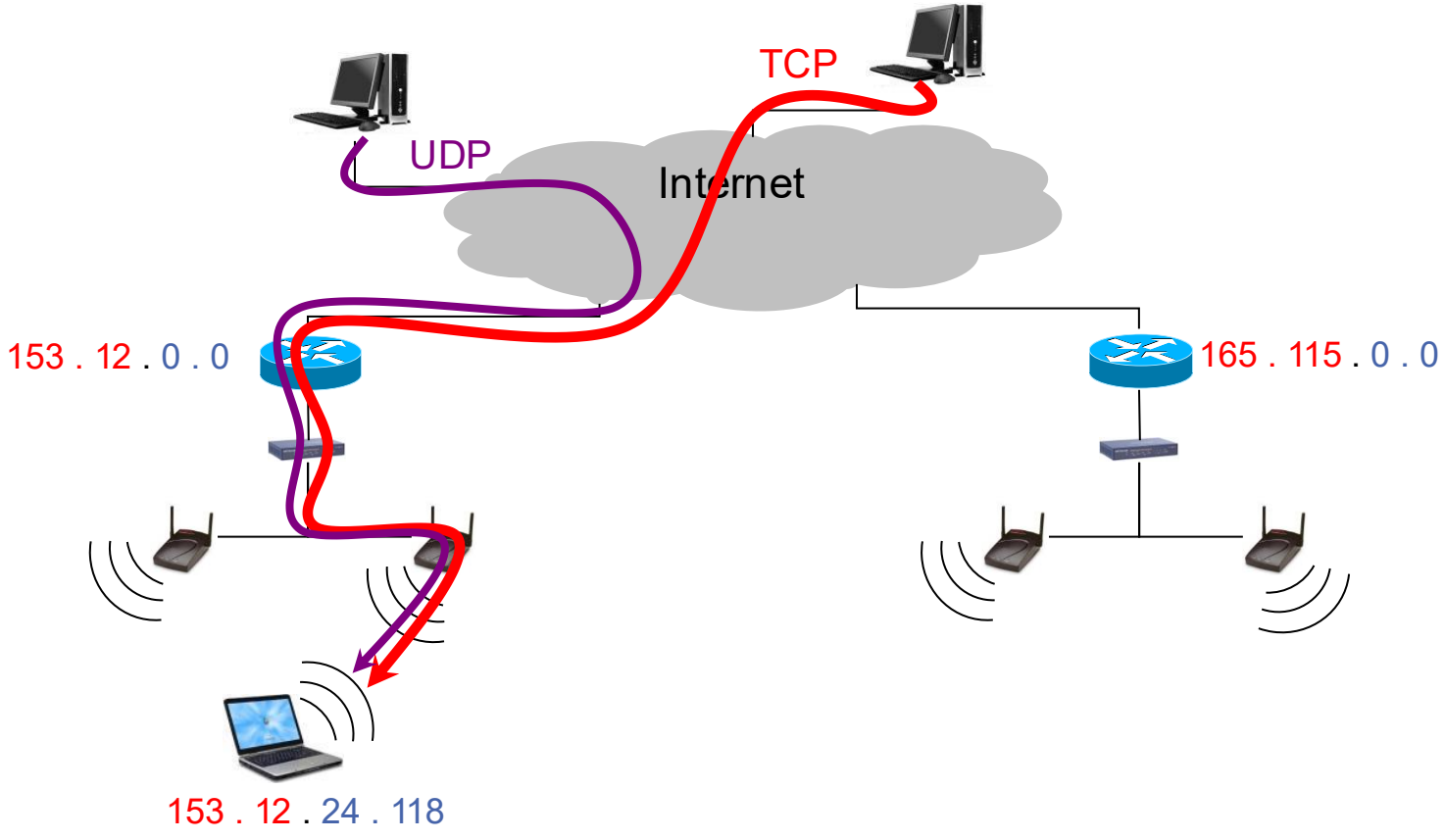
- Scanning is the main reason for latency
  - Approx. 90% of the deceleration
  - Approx. 80% of the messages exchanged (depending on mobility and quality of the radio connection)



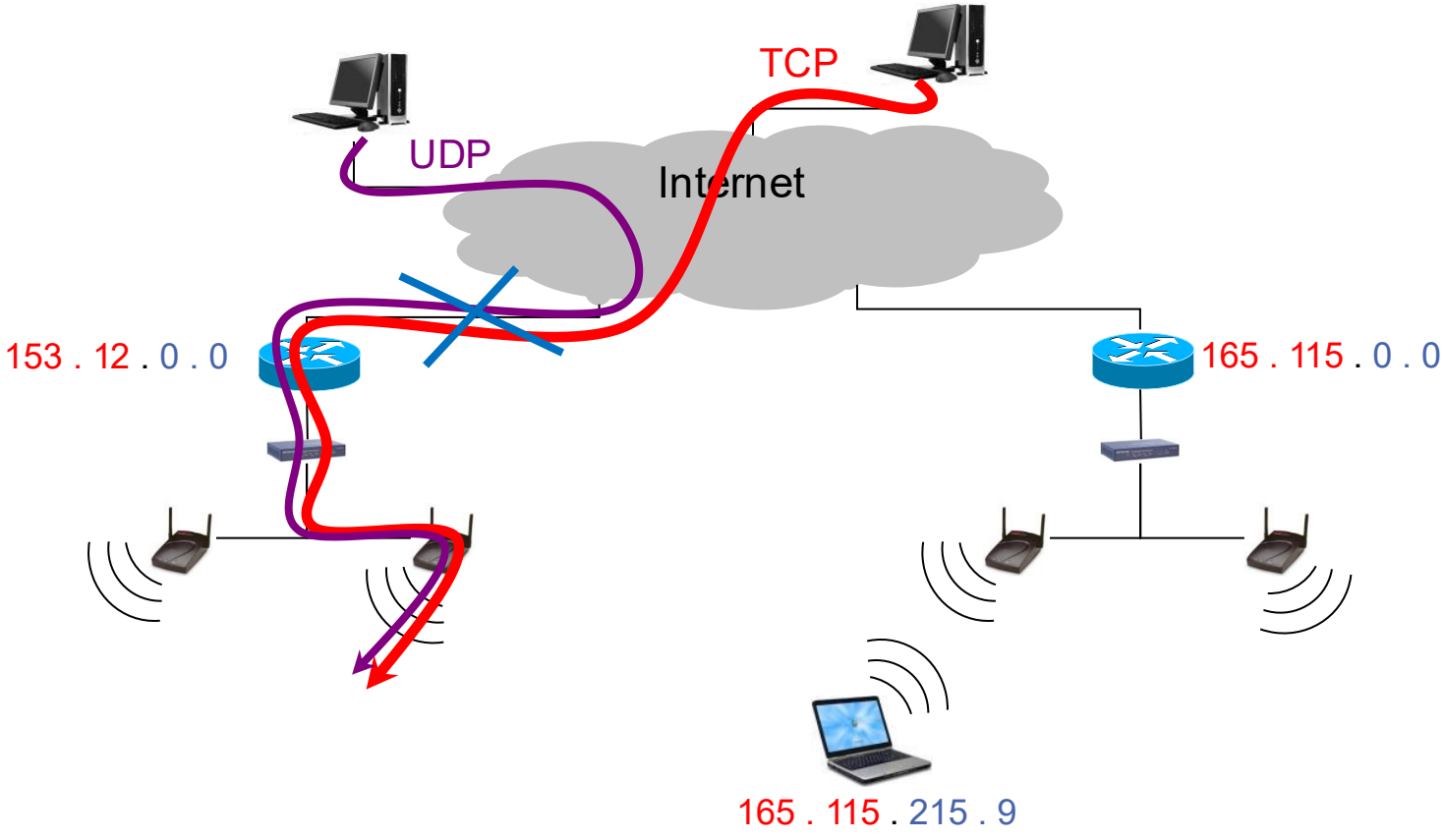
(Re-)Routing

(Re-)Routing: Problems

- IP addresses used for two purposes:
  - Locator: Specifies path through the network
  - Identifier: Specifies transport layer endpoint
- Problem: We cannot change locator while preserving identifier!



(Re-)Routing: Problems



Disconnection should be minimized / avoided!

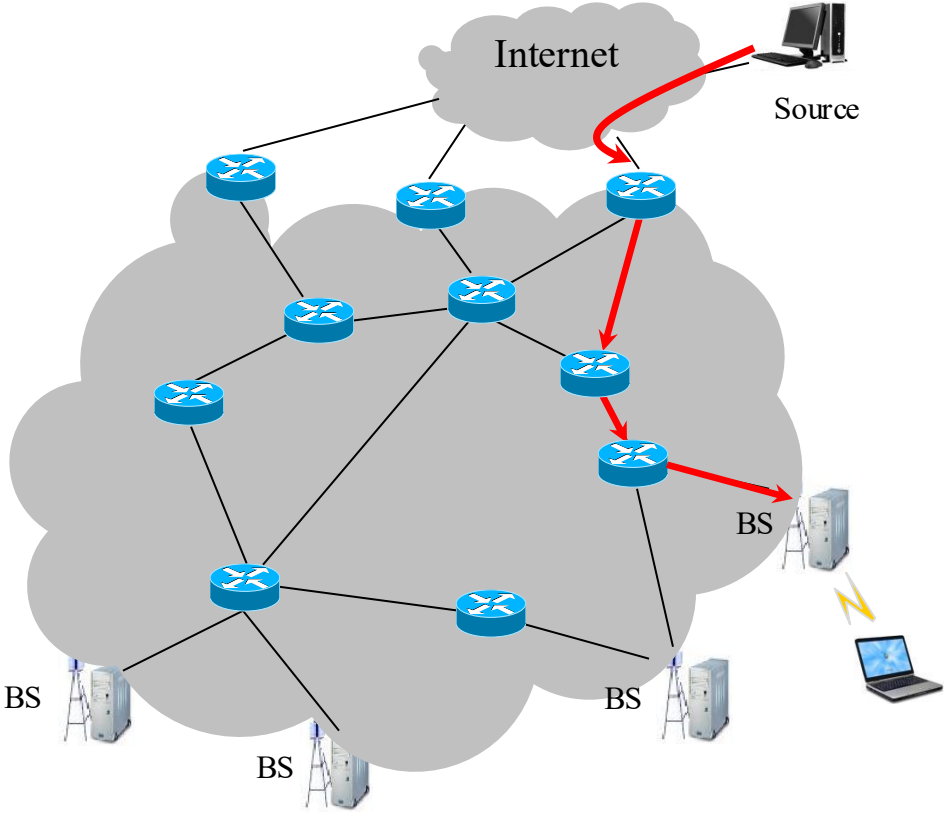
Solutions

- Solution requirements
  - Compatibility: No changes to (fixed) devices required
  - Transparency: Invisible to higher protocol layers
  - Scalable and efficient
  - Security

- Solutions
  - Specific routes per device
  - Modification of the destination address
  - Tunneling

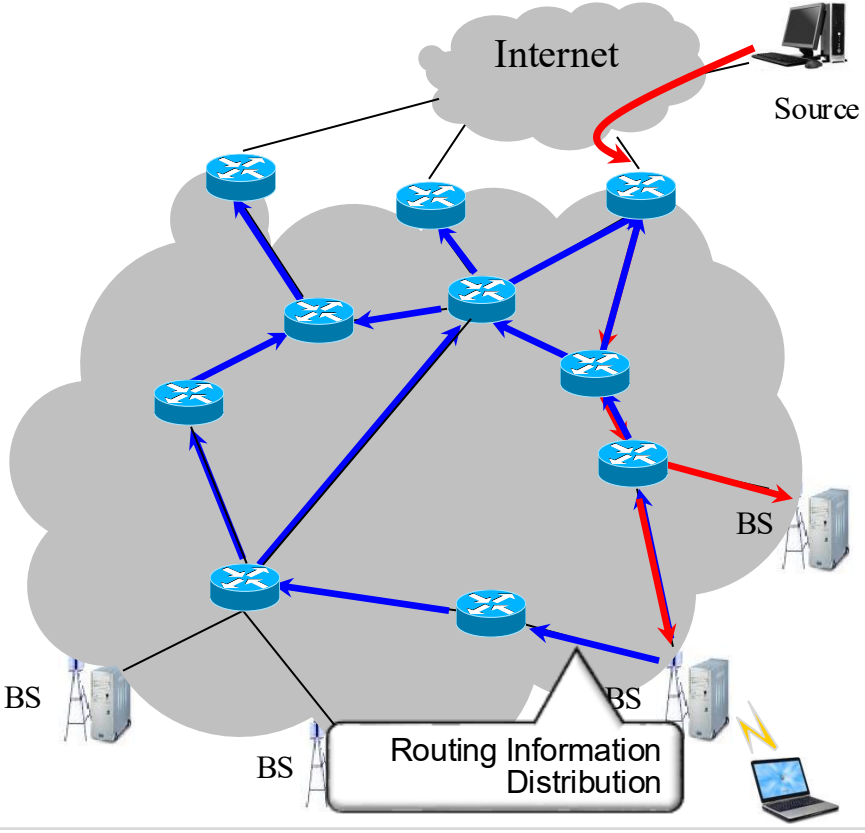
Specific routes per device

- Modification of routing table entries for correct forwarding
- Routing information for any device, anywhere



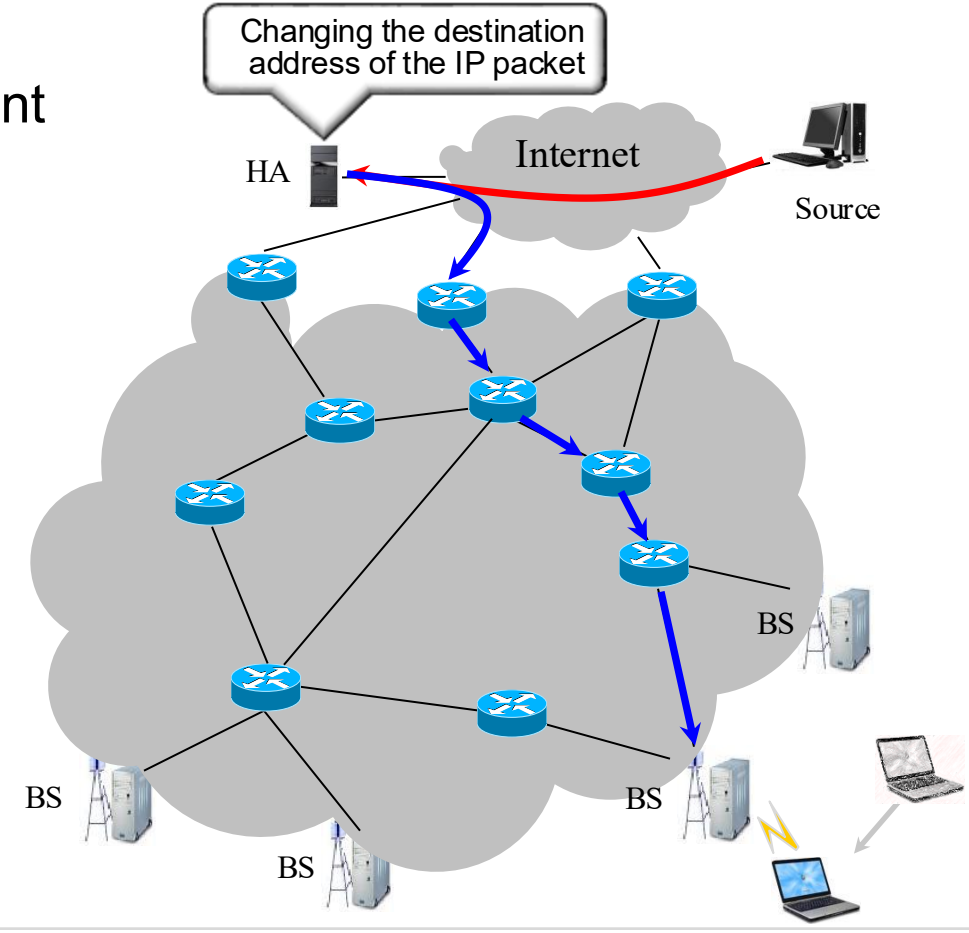
Specific routes per device

- Modification of routing table entries for correct forwarding
- Routing information for any device, anywhere
  - Does not scale for large networks
  - Security issues caused by updates to the routing table



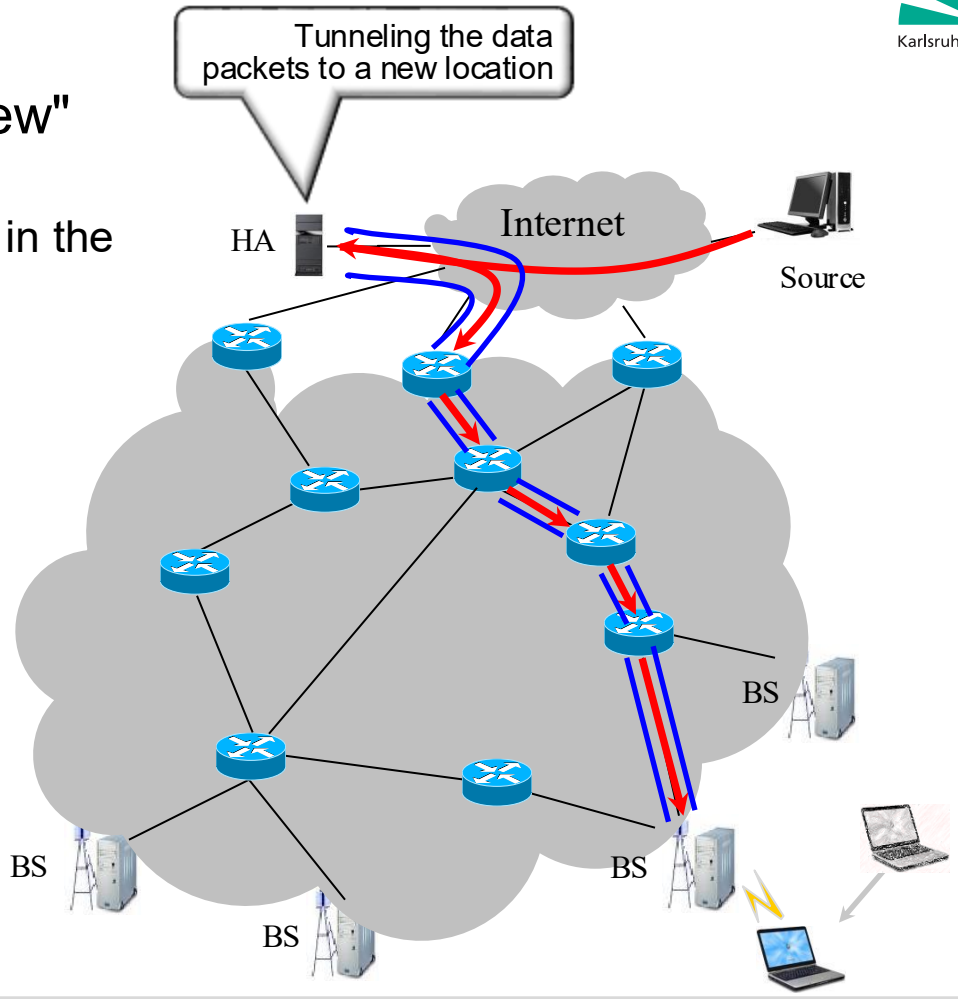
Modification of destination address

- Adjustment of the destination address according to the current position
  - Requires Mobility Anchor (Home Agent)
  - Problems due to address changes on higher shifts
  - Security



Tunneling

- Forwarding the packets in a "new" packet
  - Source and destination addresses in the "inner" package are retained
    - Transparent for higher layers
  - Still requires mobility anchor (Home Agent)
  - Security



Summary

- Communication with mobile device required
  - Position Management
    - Where is the mobile device (connected to the network)?
    - Trade-off between search effort and update effort
    - Hierarchy of Registers and Positioning Areas
  - Handover
    - Change to "better" cell
    - E.g. WLAN: Finding the better cell through active / passive scanning
  - (Re-)Routing
    - Data packets must reach mobile device
    - Difficult due to the dual function of IP addresses
    - Changing routing information in the network, changing the destination address in the packet, tunneling

Exercises - Mobility Management

4.1 What must be ensured in any case to ensure communication with a mobile device?

4.2 How can outdated position data be dealt with in HLR?

4.3 How is the "Single Point of Failure" problem countered in the case of LR?

4.4 What are the main reasons for handover between cells?

4.5 What are the advantages and disadvantages of active and passive scanning?

4.6 Why is mobility problematic for established data connections (at the transport layer)?

4.7 What are the solutions for Internet Routing mobility?

References and further reading

[4.1] K. Rothermel, B. Koldehofe, Lecture Mobile Computing SS 2011, http://www.ipvs.uni-stuttgart.de/abteilungen/vs/lehre/lehrveranstaltungen/vorlesungen/SS11/Mobile_Computing

[4.2] J. Kurose and K. Ross: Computer Networking – A Top-Down Approach. 7th ed., Pearson, 2016. (Kapitel 6)